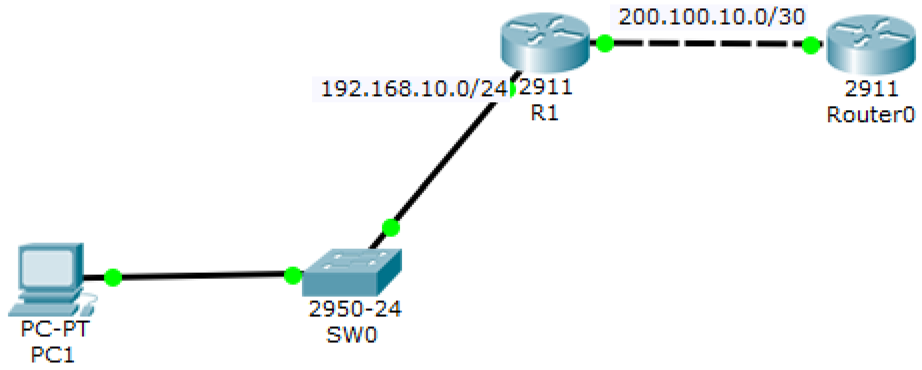
I3304 Computer Networks Administration and Security

Lab 9 – SNMP

Objective: Query the MIB of a switch and a router

1 Topology

We build a very simple topology so that the PC, which has a MIB browser, can navigate and view the MIB elements of a switch and a router.



IP Address Configuration:

- **PC 1:** 192.168.10.2/24
- **R1:**
  - GigabitEthernet0/0: 200.100.10.1/30
  - GigabitEthernet0/1: 192.168.10.1/24
- **R2:** GigabitEthernet0/0: 200.100.10.2/30

SW0 Configuration:

```
Switch>enable
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int vlan 1
Switch(config-if)#ip address 192.168.10.4 255.255.255.0
Switch(config-if)#no shutdown
```

2 SNMP Agent

To enable the SNMP agent, specify the community and the authorized access (RO for read-only and RW for read-write).

The command is exactly the same on the switch and the router:

```
R1> enable
R1# configure terminal
```

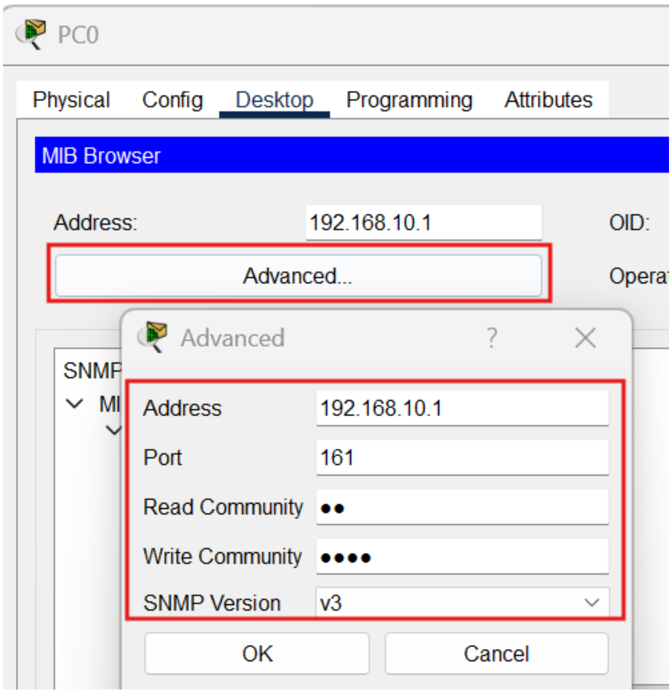
```
Enter configuration commands, one per line. End with CNTL/Z.

R1(config)# snmp-server community R1 ro
R1(config)# snmp-server community R1rw rw
R1(config)#
```

In Packet Tracer, this is the only available command. In real equipment, there are many more (SNMP traps or information).

3 Browsing the MIBs

On the PC, under the **Desktop** tab, you will find the **MIB Browser** tool:



By clicking **Advanced...**, you can define the IP address of the agent to query and the community name.

A. Enter the information as shown in the screenshot:

- **Address:** 192.168.10.1 (This is the IP address of R1)
- **Read Community:** R1 (from the "read only (ro) community name")
- **Write Community:** R1rw (from the "read and write (rw) community")
- **SNMP version**, choose **V3** and click **OK**.

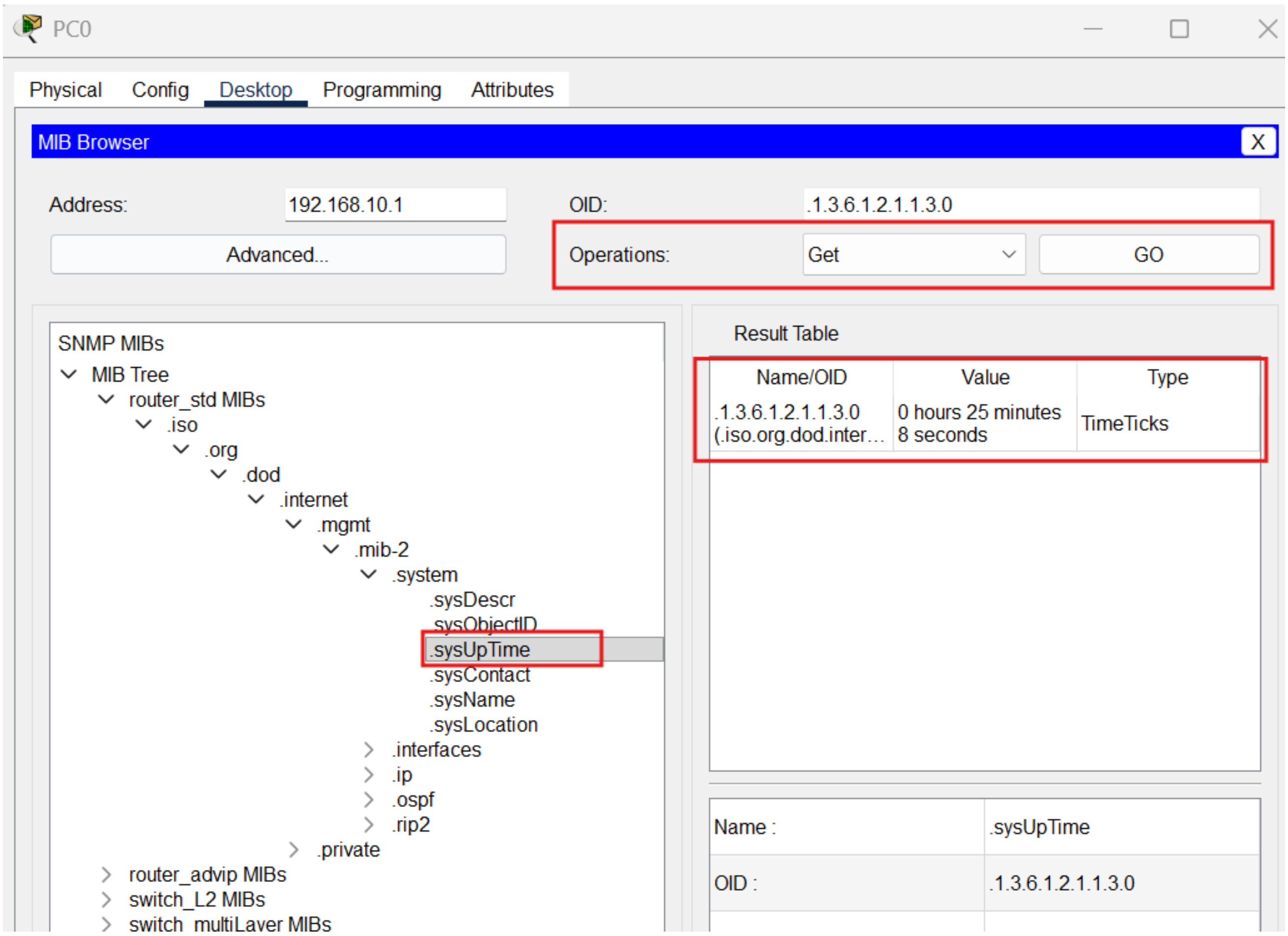
For example, we enter the IP address of router R1. We will then browse the switch's MIB (in reality, the PC must have the MIB file to identify the element to query).

B. Example: Getting the System Uptime

To retrieve the **system uptime** (how long the router has been running):

- Go to the **"router_std"** MIB
- Find the required element

- Check that **Operation** is set to **Get**
- Click **GO**



You will see the **OID** (Object Identifier), which uniquely identifies any node in the MIB tree. On the right side, you can also see that MIBs are specific to each type of device.

C. Tasks

Expand the MIB tree on the left side of the application and select the node:

MIB Tree.router_std MIBs.iso.org.dod.internet.mgmt.mib-2.system.sysDescr

- Write down the **OID value**.
- Which **IOS version** is running?
- Verify this directly on the router using the CLI command: **show version**

Now select the node:

MIB Tree.router_std MIBs.iso.org.dod.internet.mgmt.mib-2.system.sysUpTime

- Write down the OID value.
- How long has the router been running?

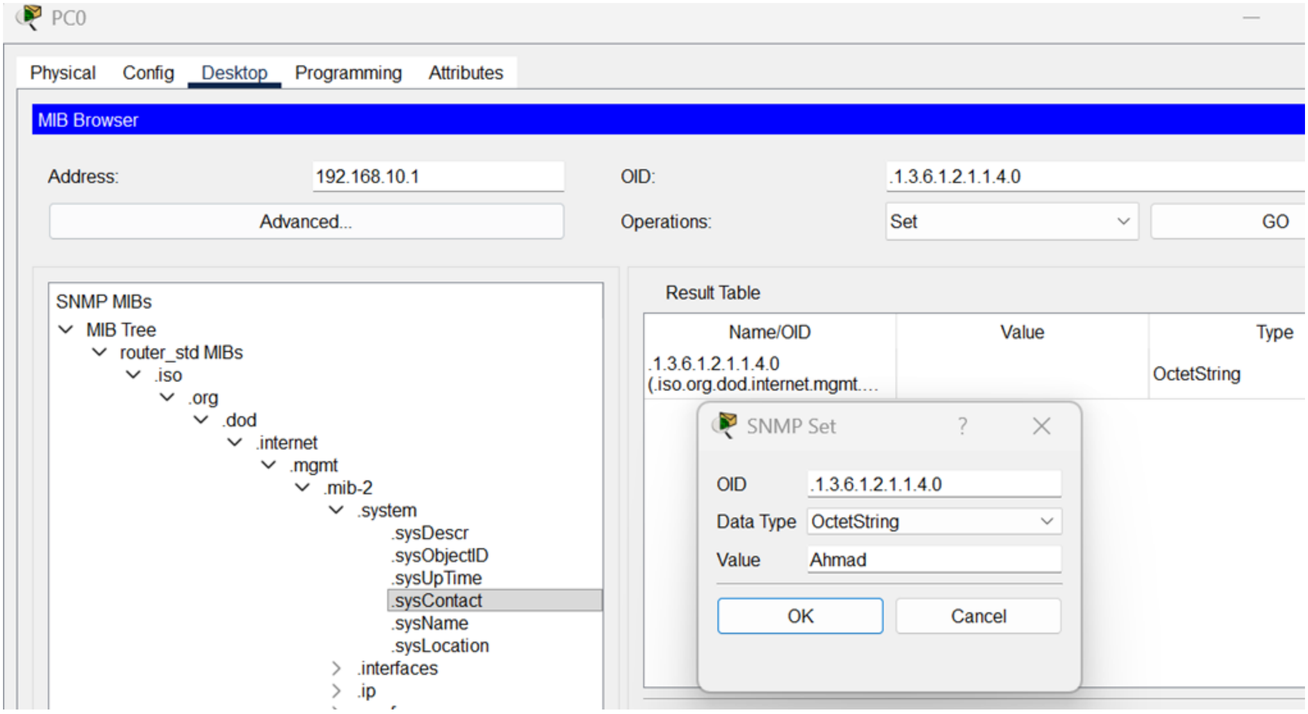
D. Modifying Values (Read-Write Variables)

Some variables can be modified via the SNMP agent. These have **read-write** access.

When a new managed device (agent) is installed, it may be necessary to add organization-specific information.

Steps:

1. Select the **sysContact** node.
2. Change the operation from **Get** to **Set**.
3. Choose data type **OctetString**.
4. Type your name and click **GO**.



Then:

- Change the operation back from **Set** to **Get**.
- Click **GO** again to verify that the data was correctly stored.

E. Additional Tasks

- Modify the value of the **sysLocation** node to **LAB** and verify that it was correctly changed.
- Enable the SNMP agent on **Router 0**. Update the MIB Browser settings on the PC (Advanced) and perform several MIB queries and value modifications, verifying that they were correctly changed.
- Enable the SNMP agent on **SW0**. Update the MIB Browser settings on the PC (Advanced) and perform several MIB queries and value modifications, verifying that they were correctly changed.